MOHAMED ABD ELRHMAN AZZAM

Data Analytics | AI / Machine Learning | AI Agents | Software & Automation

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LinkedIn | GitHub

PROFILE

M.Sc. student in Autonomy Technologies at Friedrich-Alexander-Universität Erlangen–Nürnberg (FAU), with an Excellent-with-Honors B.Sc. in Computer and Systems Engineering (top 5%). Builds data-driven applications, AI agents, LLM workflows and automation systems using Python, SQL, Power BI and modern web technologies. Experience includes LoRA fine-tuning, retrieval-augmented generation (RAG), BI dashboards, REST API integrations, and automated testing. Combines analytical problem solving with hands-on prototyping, documentation and technical communication.

TECHNICAL SKILLS

- **Data Analytics & BI:** Python, Pandas, NumPy, SQL, PostgreSQL, Power BI, DAX, Excel, Matplotlib, data cleaning, exploratory analysis, visualization, star schemas, Row-Level Security.
- **AI / ML & LLMs:** PyTorch, TensorFlow, Keras, scikit-learn, LangChain, AI agents, RAG, ChromaDB, FAISS, Transformers, Qwen, LoRA, prompt engineering, NLP, computer vision, model evaluation.
- **Software & Testing:** Python OOP, C++, FastAPI, REST APIs, JSON, JavaScript, TypeScript, React, Next.js, HTML, CSS, Git/GitHub, Playwright, Vitest, Selenium, Beautiful Soup.
- **Automation & Infrastructure:** n8n, Power Automate, Power Apps, AWS EC2, Docker, Linux, Firebase, GitHub Copilot Agent Mode, GitHub Spec Kit, VS Code, Jupyter.
- **Business & Communication:** Requirements analysis, prototype development, research, technical documentation, workshop facilitation, teaching; Microsoft Word, Excel and PowerPoint (intermediate).

SELECTED PROJECTS

Data Governance & GDPR Compliance Dashboard — GitHub

Power BI, DAX, Python, Pandas, Star Schema, Row-Level Security

- Developed a BI dashboard for a **simulated healthcare-equipment CRM**, analyzing data quality, consent, contracts and access across three regions; created synthetic datasets with Python.
- Implemented **25+ DAX measures** and row-level security roles to surface consent gaps, contract expirations and high-risk access patterns.

JobHunter – AI-Assisted Job Discovery & Evaluation — GitHub

Python, JobSpy, LLMs, JSON, Docker, Telegram Bot API

- Built a Python workflow for job discovery and deduplication, five-dimension AI-based role evaluation, structured reporting, and tailored application document generation.
- Applied German-language requirement filtering and deterministic JSON-patch CV edits to maintain consistency in generated documents.

Sanad – Tool-Using AI Agent for Case Analysis — GitHub

Python, LangChain, RAG, ChromaDB, Pydantic, Streamlit

- Built a charity case-analysis assistant combining semantic search, structured income/expense extraction and a code-driven calculator tool for reliable financial queries.
- Used retrieval against support guidelines to aid case evaluation; recorded **50% less manual guideline-evaluation time** in the project.

Turjuman – Fine-Tuned English-to-Arabic LLM — GitHub

Qwen-1.5B, PyTorch, Transformers, LoRA, LLaMA-Factory, W&B

- Fine-tuned Qwen-1.5B with LoRA using a curated bilingual dataset; monitored training and validation metrics.
- Produced 3,000+ structured examples from Wikipedia passages and GPT-4o-mini-assisted knowledge distillation, focusing on context-sensitive English-to-Arabic translation.

ERP-Yarn – AI-Assisted Business Application — GitHub

Next.js, React, TypeScript, PostgreSQL, Drizzle, Vitest

- Developing an ERP for yarn companies, translating inventory, procurement, sales and production requirements into application logic and database workflows.
- Iteratively reviewing database operations, business rules and AI-assisted code using tests and technical feedback.

SELECTED PROJECTS (CONTINUED)

Transcript Extension – Browser Language-Learning Tools GitHub

JavaScript, Chrome Manifest V3, Python, Chrome DevTools MCP

- Built caption retrieval and local transcription workflows, bilingual subtitles, caching and diagnostics; evaluated behavior through browser UI checks and regression tests.

German Vocabulary Learning Web App GitHub

JavaScript, HTML/CSS, Firebase, Playwright

- Developed level-based vocabulary decks, flashcards, quizzes and progress tracking with Firebase authentication and cloud synchronization.
- Used Playwright to test browser workflows and account-data isolation during iterative improvements.

AI Customer Service Agent – Automated Support

n8n, RAG, Reranking, PostgreSQL, Google Workspace, AWS EC2

- Designed an AI-assisted customer-support workflow with RAG and reranking for knowledge-based interactions.
- Integrated PostgreSQL interaction logs and Google Workspace archiving on AWS EC2.

Automatic Image Colorization – U-Net

PyTorch, U-Net / Convolutional Autoencoder, LAB Color Space

- Implemented a PyTorch image-colorization network that predicts LAB color channels from grayscale image input.

EXPERIENCE

Freelance AI & Web Developer **Jan 2025 – Apr 2026**

Self-Employed, Remote

- Translated client requirements into **5+ functional proof-of-concept applications**; developed and deployed web platforms using HTML, CSS and JavaScript.
- Built AWS EC2 data pipelines integrating multiple REST APIs, reducing manual data-entry work by **60%**.

Machine Learning Trainee (Internship) **Apr 2024 – Nov 2024**

Digital Egypt Pioneers Initiative (DEPI), Mansoura

- Completed a six-month ML training program covering EDA, supervised/unsupervised learning, deep learning, NLP, generative AI and computer vision.

AI Trainee (Internship) **Aug 2023 – Sep 2023**

National Telecommunication Institute (NTI), Mansoura

- Trained in machine learning, deep learning and Microsoft Azure.

Instructor (Part-Time) **Jul 2022 – Oct 2022**

High Hopes Academy, El-Mahala El-Kubra

- Supported students through individual/group instruction, course-material organization and collaborative teaching approaches.

AI Educator & Workshop Facilitator **Feb 2020 – Feb 2025**

Life Makers Charity, Egypt

- Delivered hands-on workshops on workflow automation and API integration, helping administrative teams understand practical process improvements.

EDUCATION

M.Sc. in Autonomy Technologies Apr 2026 – Oct 2028 (Expected)
Friedrich-Alexander-Universität Erlangen–Nürnberg (FAU), Germany

B.Sc. in Computer and Systems Engineering Sep 2020 – Jul 2025
Mansoura University, Egypt – Excellent with Honors; top 5% of class.
Coursework: Machine Learning, Neural Networks, Probability & Statistics, Linear Algebra, Algorithms & Data Structures.

CERTIFICATIONS & TRAINING

Machine Learning Specialization and Deep Learning Specialization – Coursera (Andrew Ng); Data Visualization & Data Cleaning – DataCamp.

LANGUAGES

Arabic: Native | English: C1 (IELTS Band 7) | German: A2 (actively improving)